

CSP Practice Questions

1. Smart Home Device Scheduling

A smart home has 5 devices that must be scheduled in 5 time slots (1–5).

Devices:

- Washing Machine (W)
- Dishwasher (D)
- Heater (H)
- Air Conditioner (A)
- Oven (O)

Constraints:

- All devices must run at different times.
- The washing machine must run before the dryer.
- The heater cannot run at time 3.
- The air conditioner must run after the heater.
- Oven must run in time slot 5.
- The dishwasher cannot be adjacent to the Washing Machine: $|D - W| \neq 1$.

2. Traffic Light Coordination

Five intersections must be assigned different signal phases (1–5) to avoid congestion.

Intersections:

I1, I2, I3, I4, I5

Constraints:

- Adjacent intersections must have different phases.
- I1 must be before I3
- I5 cannot be phase 1:
- I2 and I4 cannot differ by more than 2 $|I2 - I4| \leq 2$
- I3 must be phase 3 $\rightarrow I3 = 3$.

3. Classroom Presentation Scheduling

Five students present in 5 time slots.

Students:

A, B, C, D, E

Constraints:

- All students present at different times $\rightarrow$ allDifferent
- A must present before B $\rightarrow A < B$
- C cannot present in slot 2 $\rightarrow C \neq 2$
- D must present immediately after C $\rightarrow D = C + 1$
- E must be in slot 5 $\rightarrow E = 5$
- B and D cannot be consecutive $\rightarrow |B - D| \neq 1$

4. Warehouse Shelf Assignment

Five products must be placed on 5 shelves.

Products:

P1, P2, P3, P4, P5

Constraints:

- a. All products on different shelves $\rightarrow \text{allDifferent}$
- b. P1 must be left of P3 $\rightarrow P1 < P3$
- c. P2 cannot be on shelf 1 $\rightarrow P2 \neq 1$
- d. P4 must be adjacent to P5 $\rightarrow |P4 - P5| = 1$
- e. P3 must be on shelf 4 $\rightarrow P3 = 4$
- f. P1 and P2 cannot be adjacent $\rightarrow |P1 - P2| \neq 1$

5. Hospital Shift Assignment

Five doctors are assigned to 5 shifts.

Doctors:

D1, D2, D3, D4, D5

Constraints:

- a. Each doctor works a different shift $\rightarrow \text{allDifferent}$
- b. D1 must work before D3 $\rightarrow D1 < D3$
- c. D2 cannot work shift 4 $\rightarrow D2 \neq 4$
- d. D4 must work immediately before D5 $\rightarrow D4 + 1 = D5$
- e. D3 must be in shift 2 or 3 $\rightarrow D3 \in \{2,3\}$
- f. D1 and D5 cannot be adjacent shifts $\rightarrow |D1 - D5| \neq 1$

6. Parking Lot Assignment

Five cars must be parked in 5 spots.

Cars:

C1, C2, C3, C4, C5

Constraints:

- Each car occupies a different spot $\rightarrow \text{allDifferent}$
- C1 must be parked before C4 $\rightarrow C1 < C4$
- C2 cannot park in spot 3 $\rightarrow C2 \neq 3$
- C3 must be adjacent to C5 $\rightarrow |C3 - C5| = 1$
- C4 must be in spot 5 $\rightarrow C4 = 5$
- C1 and C2 cannot be adjacent $\rightarrow |C1 - C2| \neq 1$

Notes for Implementation

Each problem can be modeled using:

- IntVar[]
- model.allDifferent(...)
- model.arithm(...)
- model.distance(...) (for absolute constraints)